

TECHNO-WELD is a proprietary alloy based on mainly zinc, with additions of Aluminium, copper, manganese, magnesium, and other metals. Joints may show a weak bond if excessive silicone is present in the base metal.

No flux

is recommended for the joint, since fluxing will remove the trace elements that align the dendrite structure before the bond is complete. Do not use a deoxidiser on this product. Pulsation resistance welding will allow a controlled heating of the joint.

Join testing

was undertaken by Reynolds Aluminium, and was tested on both 3003-H14 and 1100-H14 Aluminium. T joints were made by using 2 sheets of 0.01" x 2 x 3" and were subjected to a continuous 5% salt spray test. No failure was observed after 500 hrs.

Corrosion resistance

is dependent upon the surrounding environment. When tested on the Eastern Seaboard, Long Island, NY, the penetration rate reached 0.000762 mm per year. Near very heavy Industrial operations, the penetration was magnified to 0.00173mm per year.

This material is expected to perform safely wherever zinc or zinc coated steel has been used in the past with success.

Several socket joints were made to simulate return bends used in practice for 3/8" dia., .89mm wall Aluminium tubing. They had a nominal 1/4" joint length. They were subjected to 720 hrs. in a controlled humidity cabinet (100% relative humidity) at 65.5°C with no visible signs of corrosion.

Endurance limit on a 1/4" section showed a limit of 8500 psi at 100 million cycles.

Lap joints were made from 2 sheets of 1/8" thickness Aluminium, 5" long and 2" wide. Using a Tinius Olsen Tensile Strength Machine, the following results were obtained:

Standard finishes: Mechanical, electrodeposited, chemical, organic, and plastic.

temperature	379 Solidus - 390 Liquidus (°C)
working temperature	392-495°C
tensile strength	Exceeds 50 K.s.i. (345 N/mm ²)
shear strength	Exceeds 50 K.s.i. (345 N/mm ²) Die cast 1/4" section = 52,000 psi Ultimate tensile strength 40,000 psi
solidification shrinkage	1.2% die design, use .3% to .6%)
thermal expansion	25°C - 105°C = .000762mm per °C (.00003" per °C)
thermal conductivity	75°C - 145°C 0.245 cal/cu.cm./°C/second
electrical conductivity	24.9% of Copper
electrical resistivity	@ 20°C = 6.8 micro-ohm - cm
impact value	Charpy 35 ft-lb
brinell hardness	97
compressive strength	Exceeds 90,000 psi
density	@ 21°C = 6.7g/cu.cm
elongation	7.5% (1/4") 3% in 2"; solidification shrinkage 0.14"/ft

Comprehensive Material Data

TECHNO-WELD LOW TEMPERATURE ALUMINIUM FUSION WELDING RODS AND MATERIAL.

Vendee and third persons assume the risk of injury directly or indirectly caused by the material if reasonable safety procedures are not followed as provided for in the data sheet, and vendor shall not be liable for such injury.

Furthermore, vendor shall not be liable for injury to vendee or third persons, directly or indirectly caused by abnormal use of the material, even if reasonable safety precautions are followed.

All persons using this product, all persons working in an area where this product is used, and all persons handling this product should be familiar with the content of this data sheet. Posting this document for employee notification is recommended by the vendor and may be required by law.

1. Hazardous Ingredients

material/component	(CAS #)	Weight (%)	HAZARD DATA	
			OSHA PEL	ACGIH TLV
zinc (Zn)	(CAS # 7440-66-6)	90-98	2 mg/m3	2 mg/m3
aluminium (Al)	(CAS # 7429-90-5)	1.0-9.0	6.5 mg/m3	0.5 mg/m3

copper (Cu)	(CAS # 7440-50-8)	0.1-5.0	30 mg/m3	niveau d'action
DUST			1.0 mg/m3(m&D)	1.0 mg/m3(m&D)
FUME			0.1mg/m3	0.2 mg/m3

2. Physical Data

material	Zn	Al	Cu
boiling point	907°C	2467°C	2595°C
specific gravity (density @ 25°C)	7.133 g/cm3	2.699 g/cm3	8.94 g/cm3
vapeur pressure	N/A	N/A	N/A
vapeur density	N/A	N/A	N/A
solubilité in h2o	N/A	N/A	N/A
% de volatiles by vol.	N/A		
evaporation rate	N/A		
appearance and odeur	Silver coloured metal with no odour		

3. Healt Hazard Information

(a) PRIMARY ROUTES OF EXPOSURE:	Ingestion of Dust, Inhalation of Dust or Fume
ZN	Chronic inhalation of Zinc Oxide dust or fume. NOTE: Zn is a high percentage of the Techno Weld formula
Al	Inhalation of fumes from melting, casting, welding or burning and dust from grinding or cutting. NOTE : Al is a low percentaga of the Techno Weld formula
Cu	Industrial exposure to copper fumes, dusts or mists result in metal fume fever with atrophic changes in nasal mucous membranes. Chronic poisoning results in Wilson's disease, characterised by a hepatic cirrhosis, brain damage, demyelination, renal disease and copper deposition in the cornea. NOTE : Cu is a low percentage of the Techno Weld formula
(B) EFFECTS OF OVEREXPOSURE:	Ingestion of Dust, Inhalation of Dust or Fume
Acute	Excessive inhalation of fumes from many metals can produce an acute reaction known as "metal fever". Symptoms consist of chills and fever (very similar to and easily confused with flu symptoms) which comes on a few hours after such an exposure. Long-term effects have not been noted.
Chronic	Fumes may cause irritation of the respiratory tract, skin and eyes.

3.1 Emergency & First Aid Treatment

Flush eyes with water at least for 15 minutes, and seek medical assistance if neccessary.

Remove excess dust from skin and wash affected area with soap and water. Seek medical assistance if neccessary.

Treat burns from molten materials as you would any other serious burns.

Inhalation: Remove from over-exposure to fresh air: Consult Physician. Keep subject warm and resting. Perform artificial respiration if breathing has stopped

ingestion: f large amounts are ingested, give 1-2 glasses of milk. Induce vomiting only if subject is fully consious andhas not convulsed. (Ingestion of significant amounts is unlikely.) All severe ingestion cases should have immediate medical aid.

4. Fire and Explosion data

Flash Point :	N/A
Auto Ignition Temperature :	N/A
Flammable Limits in air (% by Vol.) :	N/A
Extinguishing Media :	Special mixtures of dry chemicals suitable for metal fume. Do not use water or moist sand. Fire fighters should wear self-contained breathing apparatus and protective clothing.
Specila Fire :	Metal products are no a fire hazard. However, dust generated from grinding especially if mixed with flammable coatings may present a fire or explosion hazard under certain conditions. At a temperature above 375°C this alloy may melt and continuous heating could produce metal vapour.

5. Reactive data

Incompatibilité	Pure Zinc	Incompatible with strong acids, sodiums of halogens
	Pure Aluminium	N/A
	Pure Copper	Incompatible with 1-bromo-2-propyne, fume is incompatible with acetylene gas, dust and mist are incompatible with acetylene gas and magnesium metal
Hazardous Decomposition	N/A	

Hazardous Does not occur
Polymerisation

6. Spills, Leaks & Disposal Methods.

Steps to be taken: No special precautions are necessary for spills of bulk material. If large quantities of dust are spilled, remove by vacuuming with approved HEPA type filtration or wet sweeping to prevent heavy concentrations of airborne dust. Clean-up personnel should wear respirators and protective clothing.

Waste disposal: Scrap metal can be reclaimed for re-use. Follow Government and Local regulations regarding disposal

7. Special Protection Information

Specific Personal Protection	Eyes	Goggles, Face Shield
	Hands	Use chemical resistant gloves to avoid repeated skin contact
	Other Clothing	Wear non-flammable protective clothing suitable for the process of melting, casting and torch welding
	Work Hygiene	Avoid ingestion of material. Wash hands and face before eating, drinking or consumption of tobacco
Other Handling and Storage Requirements	Use good housekeeping practices to prevent accumulations of dust and to keep airborne dust concentrations at a minimum. Avoid breathing dust or fumes. Store material away from incompatible materials, and keep dust away from sources of ignition.	